

Brew Guide

Execution guide | copy → paste → run | target: 10-30 min | human review is required

RULES — read before starting

dart analyze after every agy step — zero errors before continuing
./start.sh always — never jaspr serve directly
flavorNotes must use .join(",") not .jointoString(",") — check every file
@css goes on MethodSelector class — not on State class
path in middleware is "api/gemini" — no leading slash
new URL() is broken in Jaspr — use window.location.pathname instead

1 — Environment (run once)

VERIFY

```
flutter --version && jaspr --version && agy --version && node --version  
# Expected: Flutter 3.44.1  Jaspr 0.23.1  agy 1.0.5  Node v26
```

AUTHENTICATE

```
agy auth  # opens browser — sign in with Google account
```

2 — Create project

TERMINAL

```
cd /Users/aiarguelles/Development  
jaspr create flutter-conf-jaspr-demo  
  
# Answers:  
# Directory:      flutter-conf-jaspr-demo  
# Mode:           server  
# Routing:        Y  
# Server routing: Y  
# Web embedding:  N  
# Flutter plugins: N  
  
cd flutter-conf-jaspr-demo
```

3 — Fix pubspec.yaml

TERMINAL

```
# Open pubspec.yaml and change:  
jaspr:  
  mode: server  # was: static — CRITICAL  
  
# Add under dependencies:
```

```
http: ^1.2.0
shelf: ^1.4.0
shelf_router: ^1.1.0

dart pub get
```

4 — API key setup

TERMINAL

```
echo "GEMINI_API_KEY=your_key_here" > .env
echo ".env" >> .gitignore
echo "start.sh" >> .gitignore

echo '#!/bin/bash' > start.sh
echo 'export GEMINI_API_KEY="your_key_here"' >> start.sh
echo 'jaspr serve' >> start.sh
chmod +x start.sh
```

5 — Create directories

TERMINAL

```
mkdir -p lib/handlers lib/services lib/specs
```

6 — Generate specs with agy

BEFORE RUNNING

Open Antigravity IDE. Grant all permissions when asked (find, read, write). Run each prompt in the Agent panel.

AGY SPEC 1 — method_selector.md

Create lib/specs/method_selector.md.
Spec for a Jaspr StatelessComponent called MethodSelector.
Four brewing method cards (Pour over, Espresso, Cold brew, French press).
Each card is a form POST to /api/gemini with a pre-defined prompt.
A text input form also POSTs to /api/gemini.
Loading overlay div#loading shown via JS onclick.
Error banner when hasError bool is true.
SCA palette from lib/constants/theme.dart.
Constraints: StatelessComponent, @css on class not State,
loading hidden by default, hasError parameter.
Output only the markdown file.

AGY SPEC 2 — coffee_detail.md

Create lib/specs/coffee_detail.md.
Spec for a Jaspr component Coffee in lib/pages/coffee.dart.
Displays a coffee recommendation from Gemini.
Constructor args: name, origin, roast, method, description, brewTime, waterTemp, grind, flavorNotes (List<String>).
Back link to / at the top.
HTML: article root, h1 name, dl/dt/dd metrics, ul/li flavorNotes.
SCA palette from lib/constants/theme.dart.
Constraints: @client, no RouteState.of(context), article as root, dl for metrics, ONE return statement only.
Output only the markdown file.

AGY SPEC 3 — gemini_service.md

Create lib/specs/gemini_service.md.
Spec for Dart server-side service lib/services/gemini_service.dart.
Returns a CoffeeBean record:
name, origin, roast, method, description, brewTime, waterTemp, grind, flavorNotes List<String> of 4 items.
Endpoint: <https://generativelanguage.googleapis.com/v1beta/models/gemini-3.5-flash:generateContent>
Auth: X-goog-api-key header — never query param.
Key from Platform.environment["GEMINI_API_KEY"].
generationConfig: maxOutputTokens 4096, responseType application/json.
System prompt constrains all fields to prevent JSON truncation.
Output only the markdown file.

7 — Generate code with agy

RULE

Run dart analyze after each step. Zero errors before continuing.

AGY CODE 1 — theme.dart

Update lib/constants/theme.dart with the SCA flavor palette.
Colors: scaCitrus #F6B36F, scaStrawberry #FF5658, scaChocolate #74301E, scaFloral #C28DBE, scaBlueberry #2659B1, scaFerment #CDAF29, scaWine #692729, scaHerb #326359, scaBay #143D37.
Neutrals: colorBackground #F5EFE4, colorSurface #FFFDF8, colorBorder #E2D5BE, colorTextDark #1C1208, colorTextMid #5C4A2A, colorTextMuted #8C7355.
Fonts: Playfair Display 500+700, DM Sans 400+500+700 (no weight 300).
h1: 3rem weight 700. p: 1rem weight 400.
Keep primaryColor = scaBlueberry.
Output only the Dart file.

VERIFY

CHECK: dart analyze — zero errors before continuing.

AGY CODE 2 — header.dart

Replace lib/components/header.dart.
"Brew Guide" wordmark only, links to /.
No Home/About navigation.
Playfair Display font, weight 500, colorTextDark.
Bottom border using colorBorder.
Use jaspr_router Link component.
Output only the Dart file.

AGY CODE 3 — gemini_service.dart — THE DEMO MOMENT

Generate lib/services/gemini_service.dart
following the spec in lib/specs/gemini_service.md exactly.
Output only the Dart file.

REVIEW REQUIRED

CHECK: File ends at the closing } of recommendCoffee function.
DELETE everything after that } if pubspec.yaml content was appended.
CHECK: generationConfig has maxOutputTokens: 4096 and responseMimeType as NESTED fields.
CHECK: Auth uses X-goog-api-key HEADER not query parameter.

AGY CODE 4 — gemini_handler.dart

Create lib/handlers/gemini_handler.dart.
Server-side Dart file — NOT compiled to browser.
Imports: dart:io, package:jaspr/server.dart, ../services/gemini_service.dart.
Function: Future<Response> handleGeminiRequest(Request request).
Read GEMINI_API_KEY from Platform.environment.
Read "preference" from POST body using Uri.splitQueryString.
Call recommendCoffee(preference, apiKey).
On success: redirect to /coffee with all CoffeeBean fields as query params.
flavorNotes: bean.flavorNotes.join(",") — NOT jointoString.
Path in redirect Uri: "/coffee" — no trailing slash.
On error: redirect to /?error=1&t=\${DateTime.now().millisecondsSinceEpoch}.
Output only the Dart file.

REVIEW REQUIRED

CHECK: flavorNotes uses .join(",") not .jointoString(",").

AGY CODE 5 — method_selector.dart

Generate lib/components/method_selector.dart
following lib/specs/method_selector.md exactly.
Additional requirements:
Each card is a <form> with hidden input pre-defined prompt:
Pour over: "Pour over coffee recommendation, light roast, floral and clean"
Espresso: "Espresso coffee recommendation, dark roast, bold and intense with crema"
Cold brew: "Cold brew coffee recommendation, smooth and low acid, refreshing"
French press: "French press coffee recommendation, medium dark roast, full body"
Every submit button onclick attribute value (copy exactly):
var loc = window.location.pathname;
history.replaceState({}, "", loc);

```
var banner = document.querySelector(".error-banner");
if(banner) banner.style.display = "none";
document.getElementById("loading").style.display="flex";
window.addEventListener("pageshow", function(e) {
  if(e.persisted) { document.getElementById("loading").style.display="none"; }
}, {once: true});
```

IMPORTANT: Do NOT use new URL(window.location.href) — Jaspr overrides the URL global and it will throw TypeError. Use window.location.pathname directly.

Error banner: div.error-banner, alignSelf center, scaWine color, const Color("#6927291a") background, shown when hasError is true.

Loading overlay: @keyframes spin on .cup, display none by default.

.gemini-input: raw {"width":"100%","box-sizing":"border-box"}.

Ask Gemini button: raw {"white-space":"nowrap"}.

.method-selector: alignItems AlignItems.stretch.

@css static getter on MethodSelector class — NOT on any State class.

Use only color constants from lib/constants/theme.dart.

Output only the Dart file.

REVIEW REQUIRED

CHECK: @css static getter is on class MethodSelector, not MethodSelectorState.

If on State: build fails with "styles getter not defined for MethodSelector".

CHECK: onclick uses window.location.pathname — NOT new URL().

AGY CODE 6 — coffee.dart

Generate lib/pages/coffee.dart

following lib/specs/coffee_detail.md exactly.

@client annotation required.

All CoffeeBean fields are final constructor arguments.

Back link above article: ☕ Brew Guide.

Wrap in div.coffee-page.

Do NOT use RouteState.of(context).

Do NOT add extension methods on BuildContext.

Build method must have exactly ONE return statement.

Use only color constants from lib/constants/theme.dart.

Output only the Dart file.

REVIEW REQUIRED

CHECK: Build method has ONE return statement only.

If two return statements exist, delete the second one and everything after it.

AGY CODE 7 — home.dart

Replace lib/pages/home.dart with a minimal wrapper.

@client annotation.

bool hasError = false constructor parameter.

Build returns MethodSelector(hasError: hasError).

Nothing else.

Output only the Dart file.

AGY CODE 8 — app.dart

Replace lib/app.dart router with two routes only:

Route 1: path "/" → Home(hasError: state.queryParams["error"] == "1")

Route 2: path "/coffee" → Coffee with all fields from state.queryParams.

flavorNotes: (state.queryParams["flavorNotes"] ?? "").split(",")

Remove /about route entirely.

Keep Header and .main div wrapper.

Path "/coffee" — no trailing slash.

Output only the Dart file.

8 — Fix main.server.dart

MANUAL EDIT

agy does not generate this file — edit manually.

lib/main.server.dart — replace entire file

```
library;

import 'package:jaspr/server.dart';
import 'app.dart';
import 'handlers/gemini_handler.dart';
import 'main.server.options.dart';

void main() {
  Jaspr.initializeApp(options: defaultServerOptions);

  ServerApp.addMiddleware((handler) {
    return (request) async {
      if (request.method == 'POST' &&
          request.url.path == 'api/gemini') { // NO leading slash
        return handleGeminiRequest(request);
      }
      return handler(request);
    };
  });

  runApp(Document(
    title: 'Brew Guide',
    styles: [],
    body: App(),
  ));
}
```

9 — Verify and run

TERMINAL

```
dart analyze
```

```
# Expected: only "info" warnings — zero errors

# If ports busy:
lsof -ti :8080 | xargs kill -9
lsof -ti :5567 | xargs kill -9

./start.sh
# Expected: Running jaspr in server rendering mode
# Open: http://localhost:8080
```

Browser verification checklist

- ☐ localhost:8080 loads — Brew Guide home visible
- ☐ DevTools Elements — shows section, h1, div.method-grid (real HTML, not canvas)
- ☐ Click a card — loading animation appears (☞ spinning)
- ☐ Coffee detail page loads with all fields
- ☐ DevTools on detail — shows article, h1, dl, ul
- ☐ Click ☞ Brew Guide — back to home works
- ☐ Go to localhost:8080/?error=1 — error banner appears centered
- ☐ Click any card from error state — banner disappears

10 — Git

```
TERMINAL
git init
git add .
git commit -m "feat: Brew Guide complete — Flutter Conf Orlando July 16"

git remote add origin https://github.com/arguellesdev/brew-guide-demo.git
git push -u origin main
```

Fallbacks — if something breaks

PORT CONFLICT

Port conflict: lsof -ti :8080 | xargs kill -9 && lsof -ti :5567 | xargs kill -9 && ./start.sh

403 ERROR

403 PERMISSION_DENIED from Gemini: server started without API key.
Stop, run ./start.sh, continue.

429 ERROR

429 quota exceeded: billing not enabled on GCP project.
Use the backend team key with billing.

JSON TRUNCATION

JSON truncation FormatException: Gemini response cut short.
Click the same card again. Model is non-deterministic — next call will work.

AGY APPENDED EXTRA CONTENT

agy appended pubspec.yaml to Dart file: delete everything after the final } of the function.

@CSS ON WRONG CLASS

dart analyze shows: styles getter not defined for MethodSelector
Fix: move @css static getter from MethodSelectorState to MethodSelector class.

DUPLICATE RETURN

Two return statements error: agy added new code but kept old code below.
Delete the second return statement and everything after it.

URL IS NOT A CONSTRUCTOR

TypeError: URL is not a constructor in browser console.
Fix: replace new URL(window.location.href) with window.location.pathname in onClickJs.

FLAVORNOTES NOT SPLIT

flavorNotes shows as one long string with no commas:
Open gemini_handler.dart and change jointoString(",") to join(",").

LOADING OVERLAY NOT SHOWING

Loading overlay never appears — card navigates directly without showing spinner.
Cause: new URL() error in onclick JS crashes silently before display="flex" runs.
Fix: use window.location.pathname in onClickJs — not new URL().